

Wsl installation

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Windows (using wsl)

What/why

In simple terms: WSL is a feature in Windows that lets you run a virtual machine for Linux directly on your Windows, as if you were on an Ubuntu, Debian, or other Linux computer. The greatest advantage is that it allows to run Linux and Windows **simultaneously**.

Why linux a better OS for development: For developers, its powerful commandline interface, package managers (like apt), customizable environment are primary draws. It streamlines software installation, scripting, and automation. What's more, **JOJ** used in ENGR1010J/1510J is run within linux environment. So if you use Windows, you may encounter *"It works on my computer! Why can't it work in JOJ?"*

Notice: wsl is only an environment, you have to install **one linux distro** (like ubuntu/debian/arch) along the way.

Requirements(wsl2)

- If your computer uses windows 10:
 - For x64 systems: Version 1903 or later, with Build 18362.1049 or later.
 - For ARM64 systems: Version 2004 or later, with Build 19041 or later.
- All versions of Windows 11 support wsl2.
- For Mac and Linux users, you don't need to install wsl.

Update your windows if it doesn't match the requirements, or move on to wsl1 section to install wsl1 instead.

Notice: VMware is not compatible with Hper-V in older versions. If you are already using VMare, check whether it meets the minimum requirements: Windows 10 20H1 build 19041.264 or later; VMware Workstation/Player 15.5.5 or later.

If not, please update / uninstall VMware and Windows / don't use wsl.

Best scenario

run

```
wsl.exe --install
```

If no error along the well and it's fast, then congratulations! You have wsl installed with ubuntu!
To install another distro, run

```
wsl.exe --list --online
```

And choose the distro listed and replace <distro> with it

```
wsl.exe --install <Distro>
```

Then run to set the default distro for wsl

```
wsl.exe --set-default <Distro>
```

Now run to enter wsl

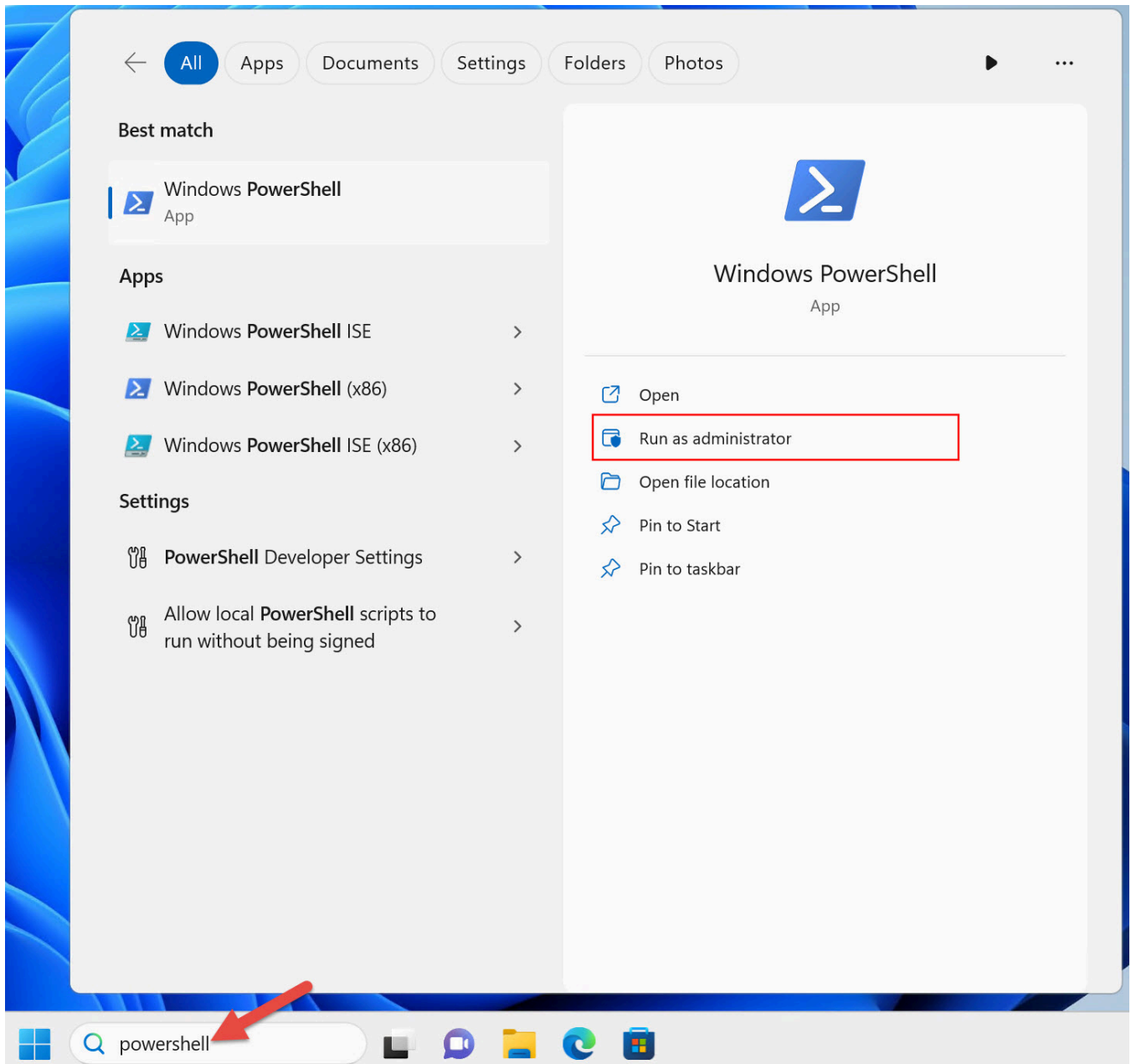
```
wsl
```

- If you are required to set user and password and no errors, then you are good to close this documentation!
- If you have no errors except that you are not required to set user and password, and when you run `wsl` you are faced with `root@xxx`, then go to `Reminder` section's Case 1.
- If you have already faced issues during `wsl --install`, then please follow `Steps` section to follow commands step by step.

Steps

There's an official tutorial: <https://learn.microsoft.com/en-us/windows/wsl/install-manual>. But you can follow our guidance as well.

1. To enable Linux feature
run `Powershell` with **administrator**



Copy and paste the following command in Powershell and hit Enter:

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
```

```
dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
```

Make sure you type exactly the same.

If you see this, then you are good to go.

```
部署映像服务和管理工具
版本: 10.0.22621.1

映像版本: 10.0.22621.2134

启用一个或多个功能
[=====100.0%=====]
操作成功完成。
```

2. Now **restart** your computer.

3. Check again whether your PC support WSL2:

- In powershell run `systeminfo`
- Scroll down to Hyper-V section
- If there are 4 "Yes" OR "A hypervisor has been detected. Features required for Hyper-V will not be displayed." ("已检测到虚拟机监控程序。将不显示 Hyper-V 所需的功能。"), then your PC is OK
- Otherwise it's not OK, please use WSL1 or other vm platform

4. Setup wsl2

```
wsl.exe --install
```

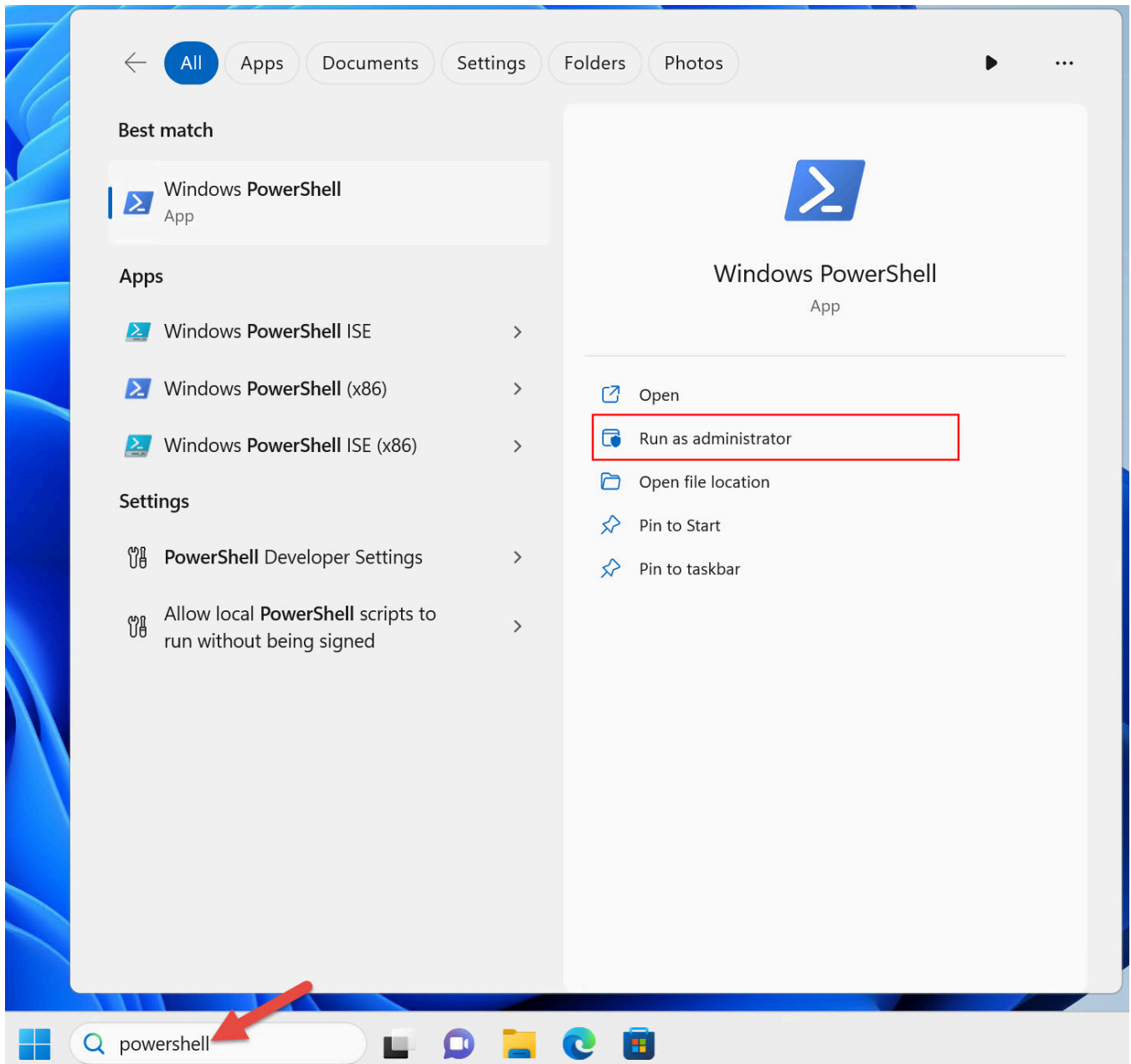
```
wsl --set-default-version 2
```

```
system32 wsl --set-default-version 2
有关与 WSL 2 的主要区别的信息, 请访问 https://aka.ms/wsl2
操作成功完成。
```

wsl1 installation

Notice: This is the fallback plan for failing to install wsl2. It's very recommended to try to install wsl2.

run `Powershell` with **administrator**



Copy and paste the following command in Powershell and hit Enter:

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
```

Now you can move on to install a linux distro: ubuntu **or** debian **or** arch.

Debug

Case 1: No error

- Go on and install linux distro in your wsl2.

Case 2: Error with link <https://aka.ms/wsl2kernel> attached

- Go to <https://aka.ms/wsl2kernel> and download a patch
- start wsl2 again
- If network is slow, consider use winget to install wsl and turn on your proxy

Case 2.1 Case 5: Error: 0x800701bc

Go to following link for tutorial:

<https://learn.microsoft.com/zh-cn/windows/wsl/install-manual#step-4---download-the-linux-kernel-update-package>

The installation link:

https://wslstorestorage.blob.core.windows.net/wslblob/wsl_update_x64.msi

Case 3: Error telling you Hyper-V is not enabled

- Go to Control Panel (控制面板) > Programs (程序) > Turn Windows features on or off (启用或关闭Windows 功能)
- Find Hyper-V

Case 3.1: No Hyper-V Settings

- You should check whether your PC support WSL2. See above.

Case 3.2: All 4 settings in Hyper-V can be ticked

- Tick all settings
- Restart your PC
- Try to start wsl2 again

Case 3.3: Some of the settings can't be ticked

This might be an issue with your hardware. Please refer to section "How to Enable Hardware Virtualization in BIOS" in <https://www.makeuseof.com/windows-11-enable-hyper-v/> and try to enable hardware virtualization. (My laptop thinkpad has this issue)

Case 4: Catastrophic failure(灾难性故障)

This error may due to corruptions during wsl installation. A reinstall may work.

You can try the following commands:

1.

```
wsl --update
```

2.

```
Disable-WindowsOptionalFeature -Online -FeatureName "Microsoft-Windows-Subsystem-Linux" -NoRestart
```

```
Disable-WindowsOptionalFeature -Online -FeatureName "VirtualMachinePlatform" -NoRestart
```

Restart your computer.

Then run these two commands again:

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
```

```
dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
```

3. go to <https://pan.sjtu.edu.cn/web/share/bc19722cabba0d5bf35826eb9e1e1b8a> and download wsl.2.6.1.0.x64.msi if your computer is x64, or wsl.2.6.1.0.arm64.msi if it's ARM.

- 4.

```
Get-AppxPackage MicrosoftCorporationII.WindowsSubsystemforLinux -AllUsers | Remove-AppxPackage
```

```
wsl --update --web-download
```

Case 5: 0x80370114Error

run

```
Disable-WindowsOptionalFeature -Online -FeatureName "HypervisorPlatform" -NoRestart
```

or

Go to **Open or close Windows features** (启用或关闭Windows功能) and cancel Windows Hypervisor Platform(windows虚拟机监控程序平台).

Then restart your computer.

Case 6: 0x8007019eError

You could try the following command:

- 1.

```
wsl --set-default-version 2
```

- 2.

```
wsl --update
```

3. try the solution of Case 2.1.

Case 7: ubuntu files not found

```
PS C:\Windows\system32> wsl list
无法将磁盘“C:\Users\10939\AppData\Local\Packages\CanonicalGroupLimited.Ubuntu24.04LTS_79rhkplfndgsc\LocalState\ext4.vhdx”附加到 WSL2: 系统找不到指定的文件。
错误代码: Wsl/Service/CreateInstance/MountDisk/HCS/ERROR_FILE_NOT_FOUND
PS C:\Windows\system32>
```

This may due to changes of the path of ubuntu files. Try:

```
wsl --unregister Ubuntu
```

And reinstall ubuntu.

Case 8: wsl command not found

Install wsl with winget:

```
winget install Microsoft.WSL
```

Otherwise

- Search online with error message on your screen (better go to stackoverflow/microsoft doc/github issues)
- Just use WSL1. It has no big issue.

Download Linux Distro

Install Linux (Ubuntu/Debian/arch, you **only** need choose **one**)

Jbox

I download latest ubuntu, debian and arch packages from ms official website, you can download **one of them**. If you use ubuntu and debian, just click the file and everything will be done automatically. If you use arch, you need to run `wsl --install --from-file <file-path>`. By arch wiki simply run `wsl --install archlinux` is fine, but it is not tested by us.

链接: <https://pan.sjtu.edu.cn/web/share/998f65b3a2da2ba1b0a3251c57e8b815>, 提取码: 8kk6
sample code(**Arch**)

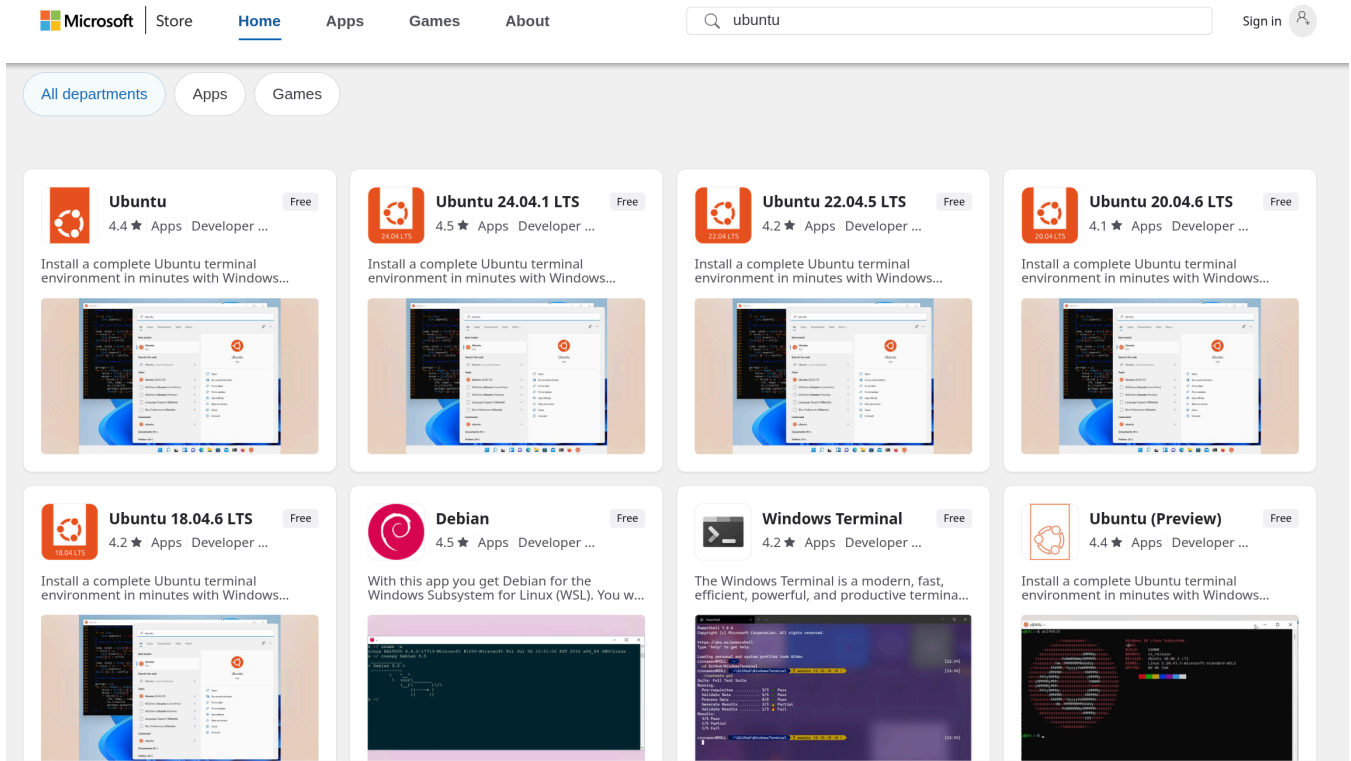
```
wsl --import Arch "$env:LOCALAPPDATA\wsl\Arch" ".\Downloads\archlinux.wsl" --version 2
```

Microsoft Store(slow and instable)

You could also download them in Microsoft Store(**no need if you successfully install them from Jbox**) but it is not recommended as it may be slow.

Take installing Ubuntu as an example:

Go to Microsoft Store and search "Ubuntu"



Download one of them. Recommend:24.04.

Hint: if it's slow, please download it from Jbox.

Ubuntu

Ubuntu is a linux distro which is widely used and simple for beginners.

Then click on **open** in microsoft store or double click the file downloaded from jbox.

Setup your user name and password. **The password will not be shown in the terminal when you type in.**

```
Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: Jaking
adduser: Please enter a username matching the regular expression configured
via the NAME_REGEX[_SYSTEM] configuration variable. Use the '--force-badname'
option to relax this check or reconfigure NAME_REGEX.
Enter new UNIX username: Jaking --force-badname
adduser: Please enter a username matching the regular expression configured
via the NAME_REGEX[_SYSTEM] configuration variable. Use the '--force-badname'
option to relax this check or reconfigure NAME_REGEX.
Enter new UNIX username: root
adduser: The user 'root' already exists.
Enter new UNIX username: [REDACTED]
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Installation successful!
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

run the following commands to install essential packages.

```
$ sudo apt update
$ sudo apt install build-essential
$ gcc --version
```

```
hydraallen@Hydraallen ~$ gcc --version
gcc (Ubuntu 11.4.0-1ubuntu1~22.04) 11.4.0
Copyright (C) 2021 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Debian

the steps are exactly the same as ubuntu.

Arch

You may refer to the official website.

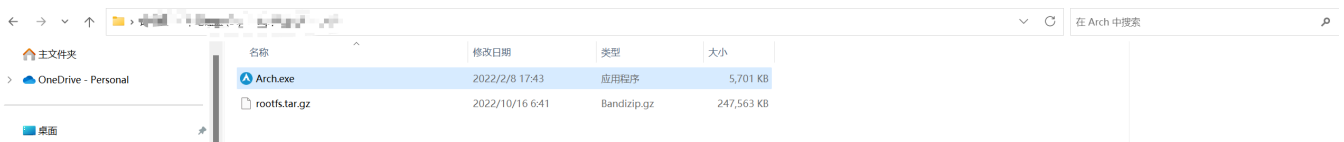
Here, we demonstrate the first method.

1. [Download](#) the installer zip.

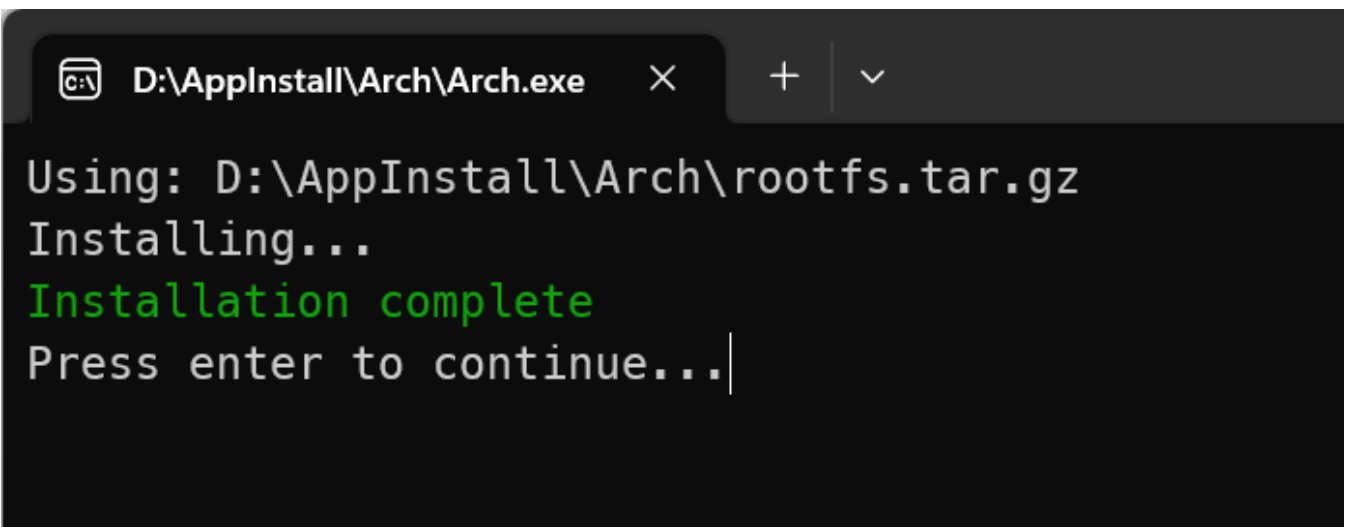
▼ Assets 8		
Arch.zip	242 MB	Oct 16, 2022
ArchWSL-AppX_22.10.16.0_x64.appx	244 MB	Oct 16, 2022
ArchWSL-AppX_22.10.16.0_x64.cer	828 Bytes	Oct 16, 2022
ArchWSL_Online-AppX_22.10.16.0_x64.appx	2.31 MB	Oct 16, 2022
ArchWSL_Online-AppX_22.10.16.0_x64.cer	828 Bytes	Oct 16, 2022
Arch_Online.zip	2.15 MB	Oct 16, 2022
Source code (zip)		Oct 16, 2022
Source code (tar.gz)		Oct 16, 2022

Hint: if it's slow, please go to [JBox](#) section and download Arch there.

2. Extract all files in zip file to the same directory. Please extract to a folder that you have write permission.
For example, `C:\Program Files` cannot be used since the rootfs cannot be modified there.

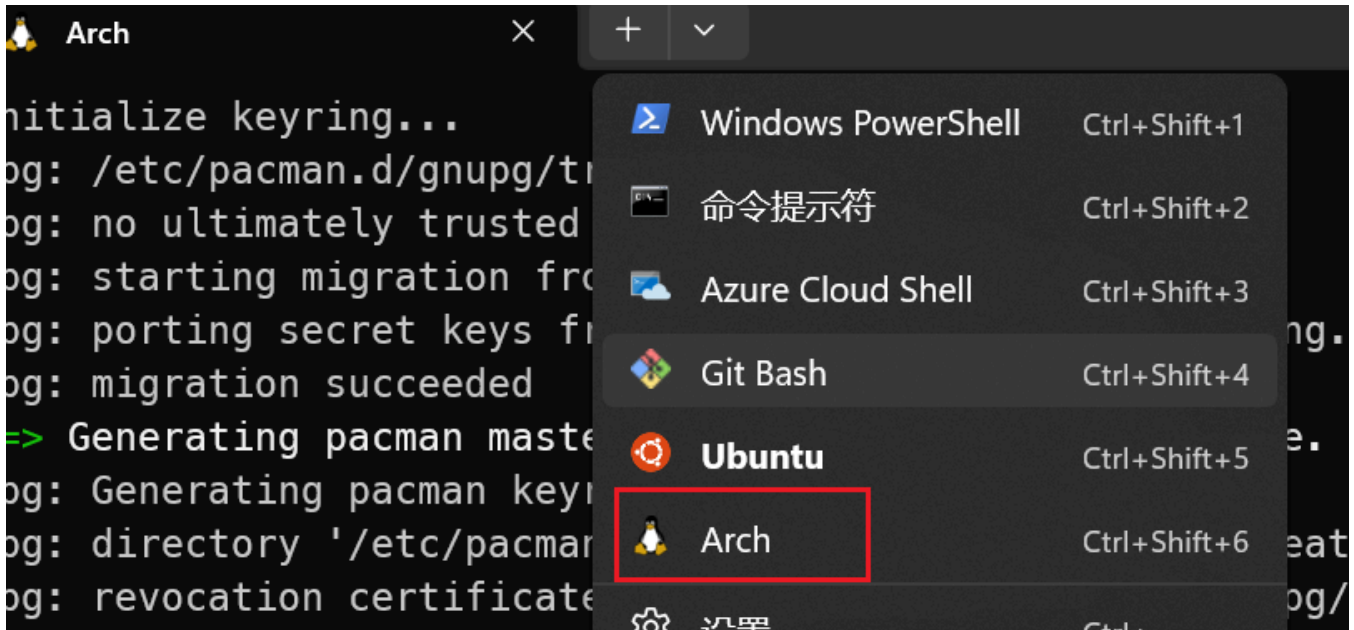


3. Run `Arch.exe` to extract the rootfs and register to WSL



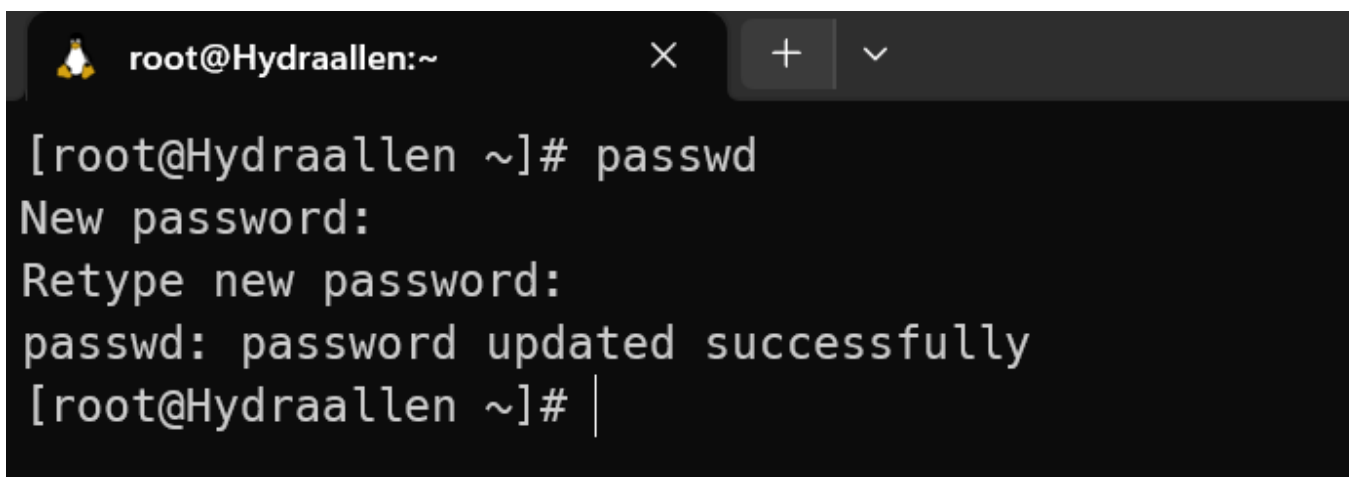
As a side note, the executable name is what is used as the WSL instance name. If you rename it, you can have multiple installs.

4. Open your terminal and choose `arch`.



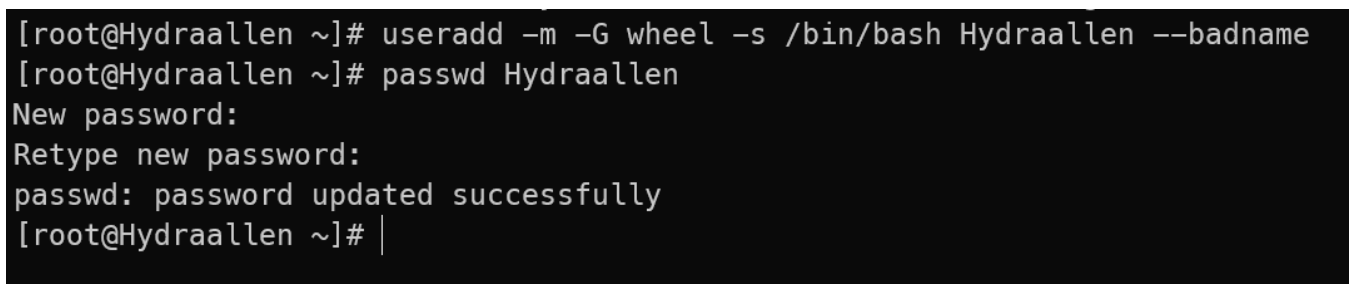
5. Setting the root password

```
passwd
```

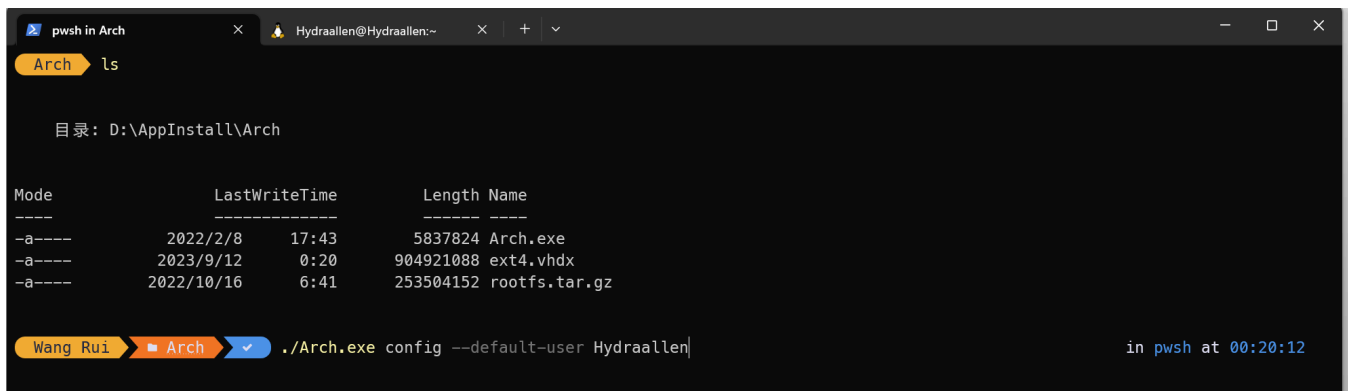


6. Set up the default user

```
$ echo "%wheel ALL=(ALL) ALL" > /etc/sudoers.d/wheel
$ useradd -m -G wheel -s /bin/bash {username}
$ passwd {username}
```



```
$ exit
$ Arch.exe config --default-user {username}
```



```
pwsh in Arch
Arch ls

目录: D:\AppInstall\Arch

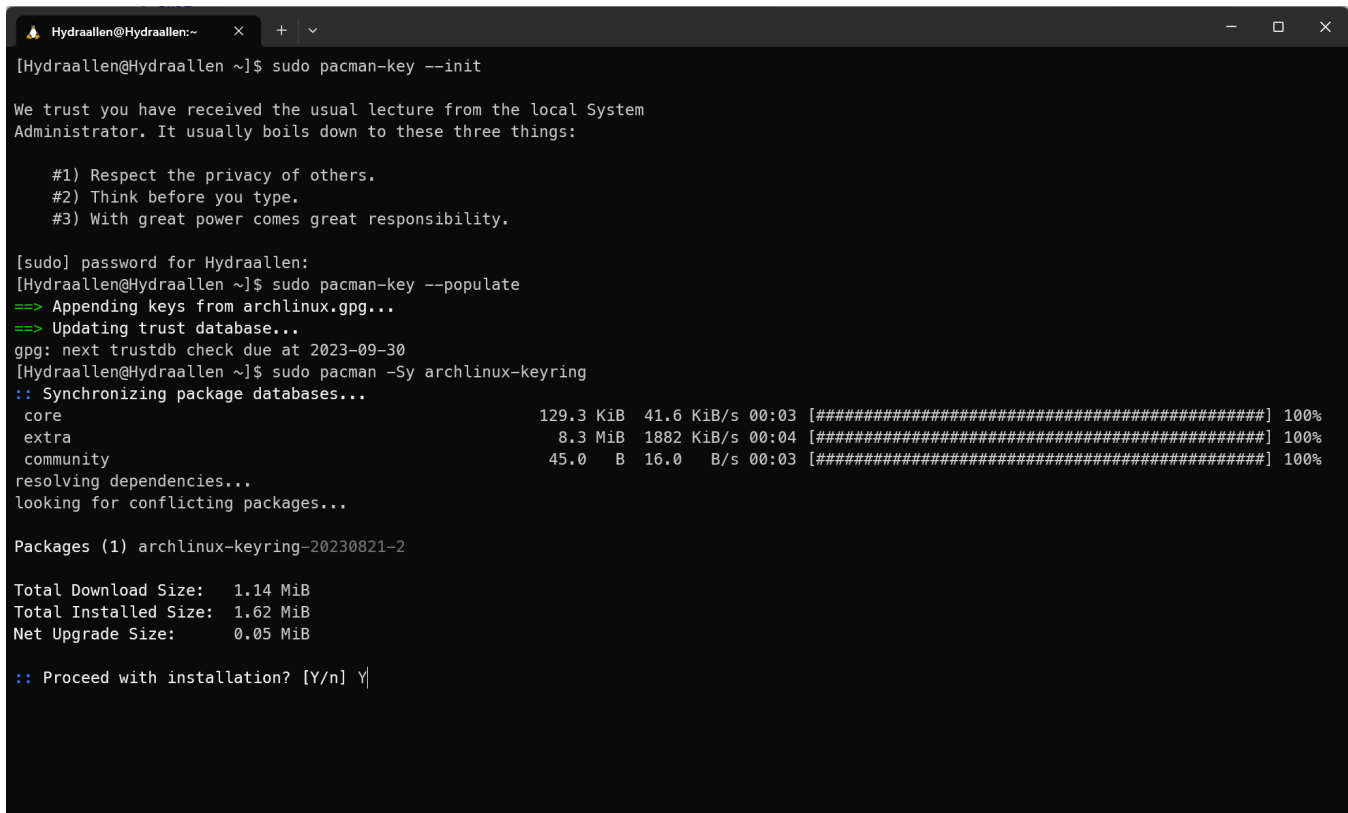
Mode                LastWriteTime         Length Name
----                -
-a----            2022/2/8          5837824 Arch.exe
-a----            2023/9/12       904921088 ext4.vhdx
-a----            2022/10/16        253504152 rootfs.tar.gz

Wang Rui Arch ./Arch.exe config --default-user Hydraallen in pwsh at 00:20:12
```

If the default user has not been changed ([issue #7](#)), please reboot the computer or alternatively, restart the LxssManager in an Admin command prompt. *Please refer to the official documentation.*

7. initialize the keyring

```
$ sudo pacman-key --init
$ sudo pacman-key --populate
$ sudo pacman -Sy archlinux-keyring
$ sudo pacman -Su
```



```
Hydraallen@Hydraallen:~$ sudo pacman-key --init
[Hydraallen@Hydraallen ~]$ sudo pacman-key --init
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for Hydraallen:
[Hydraallen@Hydraallen ~]$ sudo pacman-key --populate
==> Appending keys from archlinux.gpg...
==> Updating trust database...
gpg: next trustdb check due at 2023-09-30
[Hydraallen@Hydraallen ~]$ sudo pacman -Sy archlinux-keyring
:: Synchronizing package databases...
core                               129.3 KiB  41.6 KiB/s  00:03 [#####] 100%
extra                             8.3 MiB  1882 KiB/s  00:04 [#####] 100%
community                       45.0 B   16.0 B/s   00:03 [#####] 100%
resolving dependencies...
looking for conflicting packages...

Packages (1) archlinux-keyring-20230821-2

Total Download Size:  1.14 MiB
Total Installed Size: 1.62 MiB
Net Upgrade Size:     0.05 MiB

:: Proceed with installation? [Y/n] Y
```

8. Install gcc

```
$ sudo pacman -S gcc
```

```
[Hydraallen@Hydraallen ~]$ sudo pacman -S gcc
resolving dependencies...
looking for conflicting packages...

Packages (5) binutils-2.41-3  jansson-2.14-2  libisl-0.26-1  libmpc-1.3.1-1  gcc-13.2.1-3

Total Download Size:    55.42 MiB
Total Installed Size:  231.10 MiB

:: Proceed with installation? [Y/n] |
```

9. test gcc

```
$ gcc --version
```

```
[Hydraallen@Hydraallen ~]$ gcc --version
gcc (GCC) 13.2.1 20230801
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

[Hydraallen@Hydraallen ~]$ |
```

Reminder:

If you face issues when installing arch, please search online or just switch to ubuntu/debian.

Reminder

Case 1: No username and password required

In this case, you may see something like `root@xxx` in the terminal, which means you are logging in as a root user(superuser). You need to set up a normal user with sudo right with following commands:

```
$ adduser <username>
$ usermod -aG sudo <username>
$ exit
```

After exiting wsl, run

```
wsl --manage <distro> --set-default-user <username>
```

change <distro> to ubuntu-24.04 if you installed ubuntu. or type `wsl --list` to see the distro name you installed and replace it.

seeing `<username>@xxx:~$` everytime you enter wsl means it's successful.

Usage

Open Powershell or cmd, type:

```
wsl
```

Then you will enter the linux environment in the terminal(wsl will choose the default linux distro). Now go on and code!